Antoine Bonnet

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Research engineer building computer-use agents. At H Company, I created the agent harness behind Holo's training data, evals and products, and co-authored Holo4, H's computer-use model.

Experience

H Company 04/2025 – present
Member of Technical Staff, Agents Paris, France

- **Agent harness.** Created H's agent harness in 08/2025; 56 engineers have since contributed. It runs data collection, evals, Holo4's RL rollouts, the Agents API, HoloTab and client deployments. Found the model lost its past reasoning between steps; restoring it gave +6 to +14 pts on CUA-Gym.
  - *Automatic harness engineering.* Took Qwen3.8-27B from 24.8% to 62% on OSWorld 2, no training: evals → agents analyze traces, propose fixes → I review, merge → evals. New memory compaction: fixed 18/19 replayed bugs.
  - *From GUI to code agents.* Designed desktop, browser and cua CLIs: code beat GUI on 6/6 benchmarks (median +5 pts), up to 4.7× fewer screenshots. Merged 7 driver fixes in trycua/cua: Qwen3.8-27B on OSWorld 64% → 75%.
- **Training data.** Holo4: designed the data collection behind ∼2/3 of SFT tokens (3.6M samples, 17 environments); shaped its action format: JSON, then tool calling, then code. Audited all 79 SFT datasets (clean tokens 40% → 88%). Holo3: wrote the configs and prompts that collected its trajectory data (48% of SFT tokens).
- **Evals and environments.** Integrated ∼20 benchmarks (OSWorld, GDPval, BrowseComp); found grading bugs in 34 of OSWorld 2's 108 tasks; published 7.2k runs on trajectories.hcompany.ai. Closed reward hacks on 3 benchmarks (reading answer files, rerunning graders) by moving graders off the agent's VM. Found SFT and RL raised false success claims (done on a failed task) from 1.4% to 9.9%.
- **Products.**
  - *Agents API.* Led engineering of H's computer-use API: Python/TS SDKs, CLI, MCP server; shipped in 10 weeks.
  - *HoloTab.* Created H's Chrome browser agent: 6k users, rated 4.9/5.
  - *HoloDesktop CLI.* Built a fully local & private desktop agent on site at NVIDIA (Santa Clara) for DGX Spark.

Meta 09/2024 – 03/2025
Research Scientist Intern, GenAI Paris, France

- **Multi-agent planner.** Built a planner that delegates to sub-agents: 52.4% on GAIA.
- **Self-training.** Fine-tuned Llama 3.1 70B on its own LLM-judged runs: GAIA 33.7% → 40.2%.

MLO & NLP Labs, EPFL 06/2023 – 09/2024
Data Lead, Meditron Lausanne, Switzerland

- Led the data pipeline for *MEDITRON-70B*, a medical Llama 2: built the pretraining corpus and released the *Clinical Guidelines* dataset. 2.2k GitHub stars; featured on the Meta AI blog.

Publications

- *Holo4* (co-author); *Holo3.1*, *Holo3*, *Holo2* (contributor), H Company, 2025–26
- *Surfer 2: Cross-Platform Computer Use Agents*, 2025 (53 authors; built the orchestrator and judges)
- *MEDITRON-70B: Scaling Medical Pretraining for LLMs*, 2023 (joint second author, equal contribution; 850+ citations)

Education

EPFL 09/2022 – 03/2025
MSc Computer Science, GPA 5.47/6.00, thesis at Meta GenAI Lausanne, Switzerland

McGill University 09/2019 – 05/2022
BSc Math and Computer Science (Joint Honours, First Class), GPA 3.92/4.00, NSERC award Montréal, Canada

Skills

Agents and ML: code agents, tool calling, multi-agent systems, LLM judges, RL rollouts, vLLM
Data: trajectory collection, SFT mixtures, mid-training corpora, dedup and decontamination, Parquet, Hugging Face
Computer use: X11, macOS Accessibility, Chrome DevTools Protocol, Playwright, KVM/Kata VMs, MCP
Engineering: Python (asyncio, Pydantic), TypeScript, React, Rust (Tauri), Kubernetes, Ray, SkyPilot, AWS, Docker